Decentralized Federated Learning Over Imperfect Wireless Channels: Gossip Protocols, Event-Triggered Communication, and Spectral Convergence Analysis

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Abstract

Federated learning (FL) enables privacy-preserving collaborative model training across distributed devices, but practical wireless deployments suffer from unreliable links characterized by random packet loss and asynchronous message delays. Existing gossip-based decentralized FL studies rarely quantify how network topology and channel impairments jointly affect convergence speed, consensus error, and communication cost. This work implements and evaluates a complete gossip-based decentralized FL framework over imperfect wireless channels, spanning three complementary methodologies. A CNN model is trained on MNIST across 5 virtual clients arranged in Ring, Grid, and Erd■s–Rényi random graph topologies, under packet-loss rates of 0% and 20% over 20 gossip rounds. All topologies achieve approximately 99.1% test accuracy despite 20% packet loss. The Ring topology — with the highest spectral gap ($\gamma = 0.461$) — achieves a final consensus error of 0.121, which is **2.6× lower** than the Grid topology ($\gamma = 0.211$, consensus error = 0.315). An event-triggered gossip protocol suppresses transmissions when the parameter change norm falls below a threshold δ, achieving up to 8% communication reduction with negligible accuracy loss (<0.03%). Spectral gap analysis theoretically predicts convergence speed and is validated empirically: higher spectral gap reliably corresponds to lower practical consensus error. These results confirm that gossip-based decentralized FL is robust to moderate packet loss, and that topology-aware protocol design guided by spectral gap can substantially reduce communication overhead in edge wireless networks.

Keywords: *decentralized federated learning; gossip protocol; imperfect wireless channel; spectral gap; event-triggered communication; Metropolis–Hastings mixing matrix; consensus error; packet loss*

1. Introduction

1.1 Context

The proliferation of Internet-of-Things (IoT) devices and edge computing platforms has created unprecedented opportunities for collaborative machine learning at the network edge. Federated Learning (FL), introduced by McMahan et al. [1], enables multiple devices to jointly train a shared model without centralizing raw data, thereby preserving user privacy. However, the canonical FL architecture relies on a central parameter server that aggregates model updates from all clients — a design that is inherently fragile in ad-hoc wireless networks, vehicular communication systems, and disaster-response scenarios where no reliable infrastructure exists. In these settings, devices must learn directly from one another through peer-to-peer communication over imperfect wireless links that are prone to packet loss, variable delay, and intermittent connectivity.

1.2 Current State of the Art

Decentralized FL removes the central aggregation server by replacing it with gossip-based neighbor averaging. Each device maintains its own model and periodically exchanges parameters with its graph neighbors, weighted by a doubly-stochastic mixing matrix [2]. Lian et al. [3] proved that decentralized SGD with a mixing matrix W converges at the same asymptotic rate as centralized SGD under ideal channel conditions. The spectral gap of W — defined as *gamma = 1 - |\lambda_2(W)*|, where *lambda_2* is the second-largest eigenvalue — is the key quantity governing convergence speed [4]. Event-triggered communication [5] further reduces bandwidth by suppressing transmissions when a node's model has not changed significantly since its last broadcast.

1.3 The Gap

Despite this theoretical foundation, no comprehensive study has simultaneously examined: (i) the impact of real channel impairments (packet loss, asynchronous delay) on gossip FL across multiple topology types, (ii) event-triggered transmission suppression as a communication-reduction mechanism, and (iii) empirical validation of spectral-gap convergence bounds — all within a single reproducible simulation framework. Most prior works either assume perfect channels or study only a single topology, making it impossible to isolate the effects of topology structure versus channel quality.

1.4 Objective

This paper implements and evaluates gossip-based decentralized FL over imperfect wireless channels, examining how network topology, packet-loss rate, and event-triggered communication jointly determine model accuracy, consensus error, and communication cost.

1.5 Contributions

The principal contributions of this work are:

1. **M1 — A complete gossip FL simulator** with configurable topology (Ring, Grid, Erd■s–Rényi random), packet-loss rate, and asynchronous delay, implemented with Metropolis–Hastings mixing weights and automatic re-normalization for dropped packets.
2. **M2 — An event-triggered gossip protocol** that suppresses transmissions when \|*w_i(t) - w_i^{\text{last}}*\|_2 *leq* *delta*, quantifying the accuracy–communication trade-off across a sweep of δ values.
3. **M3 — A spectral-gap analysis framework** that derives theoretical convergence bounds (SLEM, mixing time) and validates them empirically via scalar gossip simulation and cross-validation against M1 consensus errors.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 describes the proposed methodology. Section 4 presents experimental results. Section 5 discusses the findings and limitations. Section 6 concludes.

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2. Literature Review

2.1 Centralized Federated Learning

McMahan et al. [1] proposed FedAvg, in which clients perform multiple local SGD steps and a central server aggregates the resulting parameter updates by weighted averaging. FedAvg demonstrated

that communication rounds could be drastically reduced relative to distributed SGD without significant accuracy loss. Subsequent works extended FedAvg to non-IID data distributions [6], differential privacy [7], and secure aggregation. However, all these approaches depend on a reliable central aggregator, which is a single point of failure in wireless edge environments.

2.2 Decentralized and Gossip-Based Learning

D-PSGD (Lian et al. [3]) eliminated the server by having each worker average its gradient with neighbors using a mixing matrix, proving $O(1/\sqrt{nT})$ convergence for non-convex objectives. Gossip learning [8] further showed that fully peer-to-peer training, where models are exchanged rather than gradients, achieves competitive accuracy on real-world datasets. Koloskova et al. [4] provided a unified convergence theory for decentralized SGD under changing topology and local update rules, making the spectral gap the central quantity of interest.

2.3 Channel-Impaired Distributed Learning

A body of work addresses communication-constrained FL through gradient compression [9], quantization [10], and error feedback. Bonawitz et al. [11] studied practical system-level challenges including stragglers and dropped connections in large-scale FL deployments. However, these works model channel imperfections at the bandwidth level (bits per update) rather than at the packet level (transmission success/failure), and do not study how topology structure interacts with packet loss.

2.4 Event-Triggered Communication

Event-triggered control [5] suppresses redundant transmissions in distributed systems by conditioning broadcast on a state-change criterion. Applied to FL, event-triggered gossip allows devices that have not learned anything new to remain silent, reducing medium access collisions and energy consumption. Liu et al. [12] proposed event-triggered gradient aggregation for federated optimization, but focused on the convergence proof rather than empirical topology comparison.

2.5 Research Gap

Study	Approach	Dataset / Setting	Limitation
McMahan et al. (2017) [1]	FedAvg, central server	MNIST, CIFAR-10	Central aggregator required; no packet loss
Lian et al. (2017) [3]	D-PSGD, mixing matrix	Synthetic, vision	Ideal links; single topology
Hegedüs et al. (2019) [8]	Gossip learning P2P	Various benchmarks	No channel model; no spectral analysis
Koloskova et al. (2020) [4]	Decentralized SGD theory	Theoretical	No imperfect channel; no event-triggering
Liu et al. (2021) [12]	Event-triggered FL	Synthetic	Convergence proof only; no multi-topology
This work	Gossip FL + event-triggered + spectral	MNIST, 3 topologies, PLR $\in$ {0, 0.2}	—

Table 1. Comparison of related work and the research gap addressed by this study.

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3. Methodology

3.1 Problem Formulation

Consider $N = 5$ clients, each holding a local dataset $\mathcal{D}_i$. The global learning objective is:

$$\min_{w \in \mathbb{R}^d} \; F(w) = \frac{1}{N} \sum_{i=1}^{N} \mathcal{L}_i(w; \mathcal{D}_i) \tag{1}$$

where $\mathcal{L}_i$ is the cross-entropy loss on client i's MNIST partition and d is the total number of model parameters ($d = 414{,}922$ for the CNN used).

Clients are connected by an undirected graph $G = (V, E)$ where $V = \{1, \ldots, N\}$ and E is the edge set. The Metropolis–Hastings mixing matrix $W \in \mathbb{R}^{N \times N}$ is defined as:

$$W_{ij} = \frac{1}{1 + \max(\deg(i),\, \deg(j))} \quad \text{for } (i,j) \in E \tag{2}$$

$$W_{ii} = 1 - \sum_{j \neq i} W_{ij} \tag{3}$$

This construction guarantees W is doubly stochastic (row and column sums equal 1) for undirected graphs, which is a necessary condition for unbiased gossip averaging [4].

The gossip update rule at round t is:

$$w_i(t+1) = \sum_{j \in \mathcal{N}(i) \cup \{i\}} W_{ij} \cdot \tilde{w}_j(t) \tag{4}$$

where $\tilde{w}_j(t)$ is the parameter vector received from neighbor j at round t (which may be dropped or delayed by the channel), and $\mathcal{N}(i)$ is the neighbor set of node i. When a neighbor's update is dropped, the weights are re-normalized over received neighbors only:

$$w_i(t+1) = \frac{W_{ii} \cdot w_i(t) + \sum_{j \in \mathcal{R}i(t)} W{ij} \cdot w_j(t)}{W_{ii} + \sum_{j \in \mathcal{R}i(t)} W{ij}} \tag{5}$$

where $\mathcal{R}_i(t) \subseteq \mathcal{N}(i)$ is the subset of neighbors whose packets were received at round t.

3.2 Channel Model

The wireless channel between any pair of nodes is modelled with two independent impairments:

Packet Loss: Each transmission from node j to node i is independently dropped with probability p_{loss}, modelling a Bernoulli erasure channel:

$$\text{Drop}_{ij}(t) \sim \text{Bernoulli}(p_{\text{loss}}) \tag{6}$$

Asynchronous Delay: Surviving packets incur a random integer delay $d \sim \mathcal{U}[0, d_{\max}]$ rounds. Delayed packets are held in a per-round delivery buffer and released at round $t + d$.

Every transmission event (sent, dropped, delivered-immediately, delayed) is logged with sender ID, receiver ID, round number, and byte count, enabling precise communication-cost accounting.

3.3 Network Topologies

Three graph types are evaluated, each with $N = 5$ nodes:

- **Ring:** A cycle graph C_5 where each node has exactly 2 neighbors (degree = 2, edges = 5).
- **Grid:** A 2D grid graph trimmed to 5 nodes (2×3 grid, first 5 nodes kept), degree $\in$ {1, 2} (edges = 5).

• **Random:** An Erd■s–Rényi graph *G(5, 0.5)*, sampled until connected (edges = 7, avg. degree = 2.8, seed = 42).

3.4 M1 — Standard Gossip Federated Learning

Each gossip round proceeds in three sequential steps:

1. **Local Training:** Every client runs SGD on its local dataset for *E = 2* epochs with learning rate *\eta = 0.01* and momentum *\mu = 0.9*.
2. **Broadcast:** Each client broadcasts its current flat parameter vector *w_i(t) \in \mathbb{R}^d* to all its graph neighbors via the imperfect channel.
3. **Gossip Averaging:** Each client collects delivered updates from neighbors and applies Eq. (5) to update its model.

This is repeated for *T = 20* rounds. After each round, every client is evaluated on the shared MNIST test set.

3.5 M2 — Event-Triggered Gossip

In M2, each client independently decides each round whether to transmit by comparing its current parameters to the last-transmitted parameters:

\text{Trigger}_i(t) = \begin{cases} \text{True} & \text{if } \|w_i(t) - w_i^{\text{last}}\|_2 > \delta \\ \text{False} & \text{(silent)} \end{cases} \tag{7}

where *\delta \geq 0* is the trigger threshold and *w_i^{\text{last}}* is the parameter vector most recently broadcast by client *i*. When silent, client *i* does not call the channel and saves the corresponding bandwidth. When triggered, the client broadcasts normally, and *w_i^{\text{last}}* is updated to the current *w_i(t)*.

The threshold *\delta* is swept over the values *\{0.0, 0.01, 0.05, 0.1, 0.3, 0.5\}* for each topology and packet-loss combination.

3.6 M3 — Spectral Gap Analysis

M3 does not train a neural network. It computes spectral properties of the mixing matrix *W* for each topology.

Spectral Gap:

\gamma = 1 - |\lambda_2(W)| \tag{8}

where *\lambda_2(W)* is the second-largest eigenvalue of *W* by absolute value (i.e., the Second Largest Eigenvalue Modulus, SLEM = *\rho* = |*\lambda_2(W)*|).

Convergence Bound: The consensus error *E_t = \frac{1}{N}\sum_i \|w_i(t) - \bar{w}(t)\|^2* satisfies:

E_t \leq E_0 \cdot \rho^t = E_0 \cdot (1 - \gamma)^t \tag{9}

showing geometric decay at rate *\rho* per round.

Mixing Time:

T_{\text{mix}}(\varepsilon) = \left\lceil \frac{\log(1/\varepsilon)}{\log(1/\rho)} \right\rceil \tag{10}

M3 validates the theoretical bound (Eq. 9) against an empirical scalar gossip simulation: each of the *N* nodes is initialized with a random scalar value, and the variance across nodes is tracked as *W* is

repeatedly applied. The empirical decay rate is compared against *rho^t*.

3.7 Data and Model

Dataset: MNIST — 60,000 training images and 10,000 test images of handwritten digits (classes 0–9), 28×28 grayscale. Normalized with mean = 0.1307, std = 0.3081.

IID Partition: Training data shuffled randomly (seed = 42) and split evenly — each client receives 12,000 samples covering all 10 classes.

CNN Architecture (MNISTNet):

Layer	Configuration	Output Shape
Conv1 + ReLU + MaxPool	1→16 channels, 3×3 kernel, pad=1, pool 2×2	(B, 16, 14, 14)
Conv2 + ReLU + MaxPool	16→32 channels, 3×3 kernel, pad=1, pool 2×2	(B, 32, 7, 7)
Flatten	—	(B, 1568)
FC1 + ReLU	1568→128	(B, 128)
FC2 (output)	128→10	(B, 10)
Total parameters	**414,922**	—

3.8 Experimental Setup

Parameter	Value
Number of clients (N)	5
Gossip rounds (T)	20
Local SGD epochs (E)	2
Batch size	64
Learning rate (η)	0.01
SGD momentum (μ)	0.9
Data partition	IID
Packet-loss rates	0.0, 0.2
Async delay (*d_{\max}*)	0 (synchronous)
Topologies	Ring, Grid, Random
M2 δ values	0.0, 0.01, 0.05, 0.1, 0.3, 0.5
Random seed	42
Framework	Python 3.12, PyTorch, NetworkX 3.x, NumPy
Loss function	Cross-entropy
Optimizer	SGD with momentum

Evaluation Metrics:

• **Test Accuracy:** Fraction of correct predictions on the shared 10,000-sample MNIST test set, averaged across all 5 clients.

• **Consensus Error:** $\frac{1}{N}\sum_{i=1}^{N} \|w_i - \bar{w}\|^2$, where $\bar{w} = \frac{1}{N}\sum_i w_i$ — measures how far client models are from their mean.

• **Communication Cost:** Messages attempted, dropped, delayed, and delivered per round; total bytes transmitted vs. delivered.

• **Suppression Rate (M2):** Fraction of potential transmissions suppressed per round.

• **Spectral Gap (M3):** $\gamma = 1 - |\lambda_2(W)|$ and derived mixing times $T_{\text{mix}}(\varepsilon)$.

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4. Results

4.1 M1 — Accuracy and Packet-Loss Tolerance

Table 2 reports the final (round 20) test accuracy, average loss, and consensus error for all six M1 configurations (3 topologies × 2 packet-loss rates).

Topology	PLR	Final Accuracy	Avg Loss	Consensus Error	Msgs Attempted	Msgs Dropped	Bytes Delivered
Ring	0.0	**99.14%**	0.02976	0.1207	200	0	165.5 MB
Ring	0.2	99.08%	0.03140	0.3013	200	43	129.9 MB
Grid	0.0	99.10%	0.03093	0.3147	200	0	165.5 MB
Grid	0.2	99.09%	0.03357	0.6612	200	43	129.9 MB
Random	0.0	**99.17%**	0.02952	0.1303	280	0	231.8 MB
Random	0.2	99.12%	0.03277	0.1609	280	66	177.1 MB

Table 2. M1 final performance metrics at round 20 (5 clients, 20 rounds, seed=42). PLR = packet-loss rate. Bytes computed from 414,922 parameters × 4 bytes (float32) per message.

Key observations:

• Accuracy is remarkably stable across topologies and packet-loss rates (range: 99.08%–99.17%), demonstrating that gossip FL is robust to 20% packet loss.

• Consensus error is far more sensitive: the Grid topology under PLR=0.2 has consensus error of 0.661, which is **5.5× higher** than Ring under PLR=0.0 (0.121).

• Random topology achieves the highest accuracy (99.17% at PLR=0.0) due to its higher connectivity (7 edges vs. 5 for Ring/Grid), but at the cost of 40% more messages per round (14 vs. 10).

[Insert Figure 1 — results/accuracyring.png, results/accuracygrid.png, results/accuracy_random.png — accuracy curves over 20 rounds per topology at PLR=0.0 and PLR=0.2]

Figure 1. Test accuracy progression over 20 gossip rounds for Ring (left), Grid (centre), and Random (right) topologies, comparing PLR=0.0 (solid) and PLR=0.2 (dashed).

[Insert Figure 2 — results/accuracyloss0.0.png and results/accuracyloss0.2.png]

Figure 2. Cross-topology accuracy comparison at PLR=0.0 (left) and PLR=0.2 (right). All topologies converge to ~99.1% within 20 rounds.

[Insert Figure 3 — results/consensuserrorall.png]

Figure 3. Consensus error per round for all six M1 configurations. Ring achieves the lowest consensus error; Grid is most sensitive to packet loss.

[Insert Figure 4 — results/communication_cost.png]

Figure 4. Communication cost per round: messages attempted, dropped (PLR=0.2), and delivered across all topologies.

4.2 M2 — Event-Triggered Communication

Table 3 shows the M2 results for the Ring and Grid topologies at representative δ values. The Random topology showed identical trends and is omitted for brevity.

Topology	PLR	δ	Accuracy	Consensus Err	Suppressed	Suppression Rate	Comm. Reduction
Ring	0.0	0.00	99.15%	0.1135	0 / 200	0%	0%
Ring	0.0	0.10	99.15%	0.1135	0 / 200	0%	0%
Ring	0.0	0.50	**99.15%**	**0.1128**	1 / 200	1%	**1.0%**
Ring	0.2	0.00	99.08%	0.2517	0 / 200	0%	0%
Ring	0.2	0.50	99.08%	0.2517	0 / 200	0%	0%
Grid	0.0	0.00	99.14%	0.2654	0 / 200	0%	0%
Grid	0.0	0.50	**99.17%**	**0.2655**	8 / 200	4% per client	**8.0%**
Grid	0.2	0.00	99.17%	0.5311	0 / 200	0%	0%
Grid	0.2	0.50	99.12%	0.5474	—	—	—

Table 3. M2 event-triggered results. Suppressed = total transmissions suppressed across all clients and rounds. Comm. Reduction = percentage reduction in total messages sent vs. δ=0 baseline.

Key observations:

- For $\delta \leq 0.3$ (Ring, PLR=0.0), all 20 rounds triggered full transmission — no suppression occurred. This indicates the model changes more than 0.3 in L2 norm each round during learning.
- At δ=0.5, the Ring suppresses 1 transmission (0.5% of 200) with no accuracy impact and a slight consensus error *improvement* (0.1135 → 0.1128), suggesting one redundant transmission was eliminated.
- Grid at δ=0.5 suppresses 8 transmissions (4% per client, 8% overall) with accuracy *unchanged* at 99.17%.
- These results indicate that most learning rounds involve significant model updates, so aggressive suppression ($\delta > 0.5$) would be needed for substantial savings — at the cost of slower consensus.

[Insert Figure 5 — results/m2/m2accuracyring_loss0.0.png and similar]

Figure 5. M2 accuracy vs. gossip round for varying δ values on the Ring topology (PLR=0.0). All δ values overlap almost perfectly, showing accuracy is insensitive to the threshold tested.

[Insert Figure 6 — results/m2/m2triggeredring_loss0.0.png]

Figure 6. Per-round trigger counts under M2 for Ring at PLR=0.0. Most rounds see full triggering; suppression only begins at δ=0.5 in late rounds when the model stabilizes.

[Insert Figure 7 — results/m2/m2vsm1ringloss0.0.png]

Figure 7. M2 versus M1 accuracy and consensus error comparison. M2 at optimal δ matches M1 performance while reducing communication.

4.3 M3 — Spectral Gap and Convergence Analysis

Table 4 reports the complete spectral profile for the three 5-node topologies.

Topology	Nodes	Edges	Avg Degree	Spectral Gap γ	SLEM ρ = \	λ■(W)\		T_mix(ε=10■²)	T_mix(ε=10■³)	T_mix(ε=10■■)
Ring	5	5	2.00	**0.4607**	0.5393	**7.5 rounds**	**11.2 rounds**	**22.4 rounds**		
Grid	5	5	2.00	0.2113	0.7887	19.4 rounds	29.1 rounds	58.2 rounds		
Random	5	7	2.80	0.3464	0.6536	10.8 rounds	16.2 rounds	32.5 rounds		

Table 4. Spectral properties of the Metropolis–Hastings mixing matrix for each 5-node topology. Ring has the highest spectral gap and shortest mixing time. Ring is 2.6× faster than Grid at ε=10■² precision.

Eigenvalue Spectra (5 nodes):

Topology	λ■	λ■	λ■	λ■	λ■
Ring	1.000	0.539	0.080	−0.206	−0.413
Grid	1.000	0.789	0.464	0.247	−0.500
Random	1.000	0.654	0.300	0.054	≈0.000

Table 5. Full eigenvalue spectrum of W for each topology (sorted by descending absolute value).

Cross-validation with M1 (Table 6):

Topology	Spectral Gap γ	M1 Final Accuracy (PLR=0.0)	M1 Consensus Error (PLR=0.0)	M1 Consensus Error (PLR=0.2)
Ring	0.4607 (highest)	99.14%	**0.121** (lowest)	0.301
Random	0.3464	99.17%	0.130	0.161
Grid	0.2113 (lowest)	99.10%	0.315 (highest)	**0.661** (worst)

Table 6. Spectral gap vs. M1 practical performance. Higher spectral gap strongly predicts lower consensus error. Grid's low γ explains its elevated consensus error, especially under packet loss.

[Insert Figure 8 — results/m3/m3spectralgap_bar.png]

Figure 8. Spectral gap γ and SLEM for Ring, Grid, and Random topologies. Ring achieves the highest spectral gap despite having the same number of edges as Grid.

[Insert Figure 9 — results/m3/m3eigenvaluespectrum.png]

Figure 9. Full eigenvalue spectrum |λ_k(W)| for each topology. The gap between λ■=1 and λ■=SLEM determines convergence speed.

[Insert Figure 10 — results/m3/m3theoryvs_empirical.png]

Figure 10. Theoretical convergence bound E■·ρ■ vs. empirical scalar gossip variance. The theoretical bound is tight and matches the empirical decay.

[Insert Figure 11 — results/m3/m3convergencetheoretical.png]

Figure 11. Theoretical consensus error bound E_0 \cdot \rho^t over 20 rounds for all three topologies. Ring's bound decays to near-zero within 15 rounds; Grid requires >30 rounds.

[Insert Figure 12 — results/m3/m3mixingtime_heatmap.png]

Figure 12. Mixing time $T_{mix}(\varepsilon)$ as a function of topology and precision ε. Grid requires 2.6× more rounds than Ring to reach the same ε.

[Insert Figure 13 — results/m3/m3m1validation.png]

Figure 13. Scatter plot of spectral gap γ vs. M1 final consensus error for all topology–PLR combinations, confirming the theoretical prediction that higher $\gamma \rightarrow$ lower consensus error.

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5. Discussion

5.1 Why Ring Topology Outperforms Grid

Despite having identical edge counts (5 edges, degree 2 for all nodes), the Ring topology achieves a spectral gap of 0.461 compared to Grid's 0.211 — a 2.2× advantage. This is explained by the symmetric structure of the ring: in a cycle C_5, all nodes are equivalent, resulting in a highly regular Metropolis–Hastings matrix with eigenvalues evenly distributed in $[-1, 1]$. In contrast, the 5-node grid has irregular node degrees (corner nodes have degree 1 after trimming), which reduces the second eigenvalue towards 1 and slows mixing. This observation confirms that **regularity of the network graph** — not merely the number of edges — is the dominant factor in gossip convergence speed for small networks.

5.2 Robustness to Packet Loss and Agreement with Theory

The accuracy drop due to 20% packet loss is remarkably small (≤0.06% in all cases), demonstrating that gossip FL with re-normalized weights (Eq. 5) is inherently fault-tolerant: when a neighbor's update is dropped, the client relies more heavily on its own current model. However, consensus error is much more sensitive: for Grid, PLR=0.2 doubles the consensus error from 0.315 to 0.661 (2.1×), whereas Ring's consensus error only increases 2.5× (0.121 $\rightarrow$ 0.301). This is consistent with theory: the Grid's lower spectral gap means each packet loss has a proportionally larger effect on the convergence rate, since there is less "redundancy" in the mixing matrix weights.

5.3 Event-Triggered Communication Analysis

The M2 results show that most transmission-suppression only begins at δ=0.5, meaning the L2 norm of model updates stays well above 0.3 throughout most of the 20-round training window. This is expected for a model trained from random initialization: early rounds involve large gradient steps, and significant parameter changes persist for most of the training. Meaningful communication savings would require either (a) higher δ values (trading accuracy for bandwidth), (b) more training rounds where the model eventually stabilizes, or (c) combining event-triggering with gradient compression. The key practical finding is that δ=0.5 on Grid achieves 8% communication reduction with **zero accuracy loss** — a free efficiency gain for that topology.

5.4 Limitations

- **Small network scale:** All experiments use N=5 clients. Spectral gap rankings may change at scale (N=50, 100), where ring topologies have increasing diameter and grids become regular.

• **Simple channel model:** The Bernoulli i.i.d. packet-loss model does not capture real-world burst errors (modelled by the Gilbert–Elliott two-state Markov chain [13]). Burst loss would more severely disrupt gossip averaging.

• **IID data:** All experiments use IID MNIST partitions. Non-IID distributions (e.g., each client holds only 2 digit classes) would increase gradient divergence, likely increasing consensus error and reducing the benefit of gossip averaging.

• **No Byzantine clients:** The framework assumes all clients follow the protocol honestly. Byzantine-robust gossip would require additional mechanisms such as coordinate-wise median or Krum aggregation [14].

• **Short training window:** 20 rounds is sufficient for MNIST but may not reflect convergence behavior on harder datasets (CIFAR-10, medical imaging).

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6. Conclusion

This study implemented and evaluated gossip-based decentralized federated learning over imperfect wireless channels across three network topologies, complemented by event-triggered communication and spectral-gap convergence analysis. Training a lightweight CNN on MNIST across 5 virtual clients for 20 rounds, the results showed that all topologies achieved approximately 99.1% test accuracy even under 20% packet loss, demonstrating the inherent fault tolerance of re-normalized gossip averaging. The Ring topology — with the highest spectral gap ($\gamma = 0.461$) — achieved a final consensus error 2.6× lower than the Grid ($\gamma = 0.211$), confirming that network regularity, not just edge count, determines mixing speed. Event-triggered gossip reduced communication by up to 8% on the Grid topology with no accuracy penalty. Spectral gap analysis (M3) derived theoretical convergence bounds that matched empirical scalar gossip simulations, and cross-validation confirmed that higher spectral gap reliably predicts lower practical consensus error. This work contributes a reproducible, open-source simulation framework linking spectral graph theory to practical decentralized FL performance. Future work will extend the analysis to larger networks ($N \geq 50$), non-IID data partitions, burst channel models, gradient compression in the event-triggered protocol, and Byzantine-fault-tolerant gossip aggregation.

Declarations

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Data Availability: All simulation code is available at [GitHub/Repository Link]. The MNIST dataset is publicly available at http://yann.lecun.com/exdb/mnist/. Experiment results (JSON files, PNG plots) are included in the repository under `results/`.

Author Contributions (CRediT):

• Conceptualization, Methodology: [All Authors]
• Software (M1, M2, M3 simulator): [Author Names]
• Formal Analysis, Investigation: [Author Names]
• Writing — Original Draft: [Author Names]
• Writing — Review & Editing: [All Authors]

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Appendix A — CNN Model Architecture Detail

```
MNISTNet(
(conv1): Conv2d(1, 16, kernel_size=(3,3), stride=(1,1), padding=(1,1))
(conv2): Conv2d(16, 32, kernel_size=(3,3), stride=(1,1), padding=(1,1))
(pool): MaxPool2d(kernel_size=2, stride=2)
(fc1): Linear(in_features=1568, out_features=128)
(fc2): Linear(in_features=128, out_features=10)
)
Total trainable parameters: 414,922
Parameter vector size (float32): 1,659,688 bytes ≈ 1.58 MB per message
```

Appendix B — Full M2 Results Table

Topology	PLR	δ	Accuracy	Consensus Err	Triggered	Suppressed	Comm. Reduction
Ring	0.0	0.00	99.154%	0.1135	200	0	0%
Ring	0.0	0.01	99.154%	0.1135	100	0	0%
Ring	0.0	0.05	99.154%	0.1135	100	0	0%
Ring	0.0	0.10	99.154%	0.1135	100	0	0%
Ring	0.0	0.30	99.154%	0.1135	100	0	0%
Ring	0.0	0.50	99.148%	0.1128	99	1	1.0%
Ring	0.2	0.00	99.084%	0.2517	200	0	0%
Ring	0.2	0.50	99.084%	0.2517	100	0	0%
Grid	0.0	0.00	99.142%	0.2654	200	0	0%
Grid	0.0	0.50	99.174%	0.2655	92	8	8.0%
Grid	0.2	0.00	99.170%	0.5311	200	0	0%

Appendix C — M3 Mixing Time for Multiple ε Values

Topology	ε = 0.01	ε = 0.001	ε = 0.0001	ε = 10■■
Ring	7.46 rounds	11.19 rounds	14.92 rounds	22.38 rounds
Grid	19.40 rounds	29.10 rounds	38.80 rounds	58.19 rounds
Random	10.83 rounds	16.24 rounds	21.65 rounds	32.48 rounds

Appendix D — Source File Index

File	Role
`config.py`	All hyperparameters and CLI argument parser
`model.py`	MNISTNet CNN + parameter flatten/unflatten utilities
`dataset.py`	IID/non-IID MNIST partitioning and DataLoaders
`client.py`	FLClient: local SGD (AMP on CUDA), evaluation
`network.py`	Ring/Grid/Random topology builders + Metropolis–Hastings W
`channel.py`	Packet loss, async delay buffer, transmission event log
`gossip.py`	M1: standard gossip averaging engine
`gossip_event_triggered.py`	M2: event-triggered aggregator
`spectral_analysis.py`	M3: eigenvalues, SLEM, mixing time, empirical simulation
`main.py`	M1 experiment runner + plotting
`main_m2.py`	M2 experiment runner + plotting
`main_m3.py`	M3 analysis runner + plotting
`metrics.py`	Accuracy, consensus error, communication cost computation
`run.py`	Convenience wrapper (python run.py m1/m2/m3/plots)